

PART 13 OF 18

AI Security Operations Center (AI SOC)

Agent Detection and Response, AI-Specific SIEM, Threat Hunting,
Forensics, Incident Response, and SOC Workflow Integration

ENTERPRISE AI CONTROL ARCHITECTURE

Implementation Guide for Production AI Systems • 2026

13.1 Evolution of the SOC for AI Agent Threats

Traditional Security Operations Centers are optimized for detecting and responding to threats targeting human users and deterministic software systems. AI agent threats require SOC evolution in three dimensions: detection capabilities (new threat patterns), analyst skills (semantic understanding of AI behavior), and response procedures (AI-specific containment and recovery). The AI SOC is not a replacement for the traditional SOC—it is an extension that handles the AI threat surface as a specialization.

AI SOC Core Requirement: SOC analysts must be able to distinguish between an AI agent making unusual-but-authorized decisions and an AI agent that is compromised or misaligned. This requires semantic understanding of agent goals, context, and decision rationale—a capability not required in traditional SOC roles.

13.2 AI-Specific SIEM Architecture

13.2.1 SIEM Data Sources for AI Agents

Data Source	Content	Integration Method	Retention
Agent Action Logs	All tool calls, API requests, file operations	OpenTelemetry -> SIEM ingest	90 days hot / 3 years cold
Policy Decision Logs	All authorization decisions with context	Policy engine -> SIEM	1 year hot / 7 years cold
Behavioral Metrics	Anomaly scores, drift metrics, risk trajectories	Analytics platform -> SIEM	1 year hot
Memory Access Logs	All memory read/write operations	Memory governor -> SIEM	90 days hot / 3 years cold
Tool Invocation Logs	Tool calls, parameters, results metadata	Tool proxy -> SIEM	90 days hot / 3 years cold
Approval Gate Events	Human approval requests, decisions, reasoning	Approval workflow -> SIEM	7 years (audit record)
Identity Events	Token issuance, delegation, revocation, expiry	Identity broker -> SIEM	3 years
Circuit Breaker Events	Breaker trips, resets, incident correlation	Runtime monitor -> SIEM	3 years

13.3 Agent Detection and Response (ADR)

13.3.1 ADR: The AI Equivalent of EDR

Endpoint Detection and Response (EDR) monitors endpoints for malicious activity and enables containment and investigation. Agent Detection and Response (ADR) is the analogous capability for AI agents: continuous monitoring, behavioral anomaly detection, automated or human-initiated containment, and forensic investigation support. ADR must be deeply integrated with the agent execution environment.

13.3.2 ADR Capabilities

Capability	Description	Implementation
Continuous Behavioral Monitoring	Real-time monitoring of all agent actions against behavioral baseline	eBPF + OTel + behavioral ML model
Anomaly Detection	Statistical and ML-based detection of behavioral anomalies	Isolation Forest / LSTM anomaly detection on action sequences
Automated Containment	Circuit breaker triggers to isolate compromised agent without human delay	Runtime kill switch + capability token revocation
Human-Triggered Isolation	SOC analyst can quarantine agent with single action	Incident response dashboard + kill switch API
Forensic Evidence Collection	Capture full agent state at time of detection	State snapshot API + reasoning trace export
Root Cause Analysis	Replay agent actions in forensic environment	Action replay system in isolated sandbox
Threat Hunting	Proactive search for indicators of compromise in agent telemetry	SIEM query interface + threat hunting playbooks
Incident Timeline	Automated reconstruction of incident event sequence	Causality graph from correlated telemetry events

13.4 AI Incident Response Playbooks

13.4.1 Incident Classification

Incident Type	Indicators	Severity	Response Time
Prompt Injection Detected	Injection pattern in action params; goal drift following external content retrieval	HIGH	15 min
Data Exfiltration Suspected	Anomalous data access + external API calls; volume spike	CRITICAL	5 min
Agent Identity Compromise	Token used from unexpected location; behavioral profile mismatch	CRITICAL	5 min

Incident Type	Indicators	Severity	Response Time
Memory Poisoning	Anomalous memory write; subsequent behavior change; new false belief detected	HIGH	30 min
Privilege Escalation	Permission request outside normal scope; new capability acquisition	HIGH	15 min
Goal Hijacking	Agent actions inconsistent with declared goal; semantic drift detected	HIGH	15 min
Circuit Breaker Trip	Automated containment triggered; root cause unknown	MEDIUM-HIGH	30 min
Policy Violation	Hard constraint violation detected; unauthorized action attempted	MEDIUM	60 min

13.5 AI Forensics Procedures

13.5.1 Evidence Collection for AI Incidents

AI incident forensics must capture evidence beyond what traditional forensics requires. The chain of custody for AI incidents includes: prompts (the 'scene of crime'), reasoning traces (the 'suspect's mental state'), tool invocations (the 'physical actions'), memory states (the 'environment before and after'), and authorization decisions (the 'access controls'). All evidence must be cryptographically verified to resist tampering claims.

Evidence Type	Collection Method	Preservation Requirement
Agent state snapshot	Runtime state capture API at time of detection	Cryptographically sealed; write-once storage
Reasoning trace export	Full chain-of-thought export for incident window	Encrypted; access-controlled; chain of custody logged
Action history	SIEM query for all agent actions during incident window	Immutable SIEM export with hash verification
Memory state before/after	Memory governor snapshot at detection time + 24h prior	Hash-verified snapshot; diff analysis
Tool call logs	Full request/response logs including parameters	Immutable log export; PII protected
Authorization decisions	Policy engine decision log for incident window	Immutable export; includes full context
Communication logs	A2A and external API communication records	Immutable export with source/dest verification

13.6 Threat Hunting for AI Agents

13.6.1 AI-Specific Threat Hunting Hypotheses

Hunt: Long-tail Injection

Search for agents that retrieved external content immediately followed by unusual action sequences. Hypothesis: indirect prompt injection caused behavior change.

Hunt: Permission Creep

Identify agents whose aggregate capability token set has grown over time without corresponding task complexity growth. Hypothesis: systematic permission accumulation.

Hunt: Memory Anomaly Clusters

Identify memory writes that cluster around specific time periods and share semantic similarity. Hypothesis: coordinated memory poisoning campaign.

Hunt: Cross-Agent Correlation

Identify statistically improbable correlations in actions across different agent instances. Hypothesis: cross-agent coordination or collusion.

Hunt: Data Staging

Identify read operations across diverse data sources that aggregate toward a common external endpoint. Hypothesis: multi-stage data exfiltration in progress.

Hunt: Goal Drift Cluster

Identify time periods where multiple agent types show simultaneous goal drift. Hypothesis: coordinated attack affecting multiple agents simultaneously.